COURSE 2 · SECTION 11 · CAPSTONE BRIEF

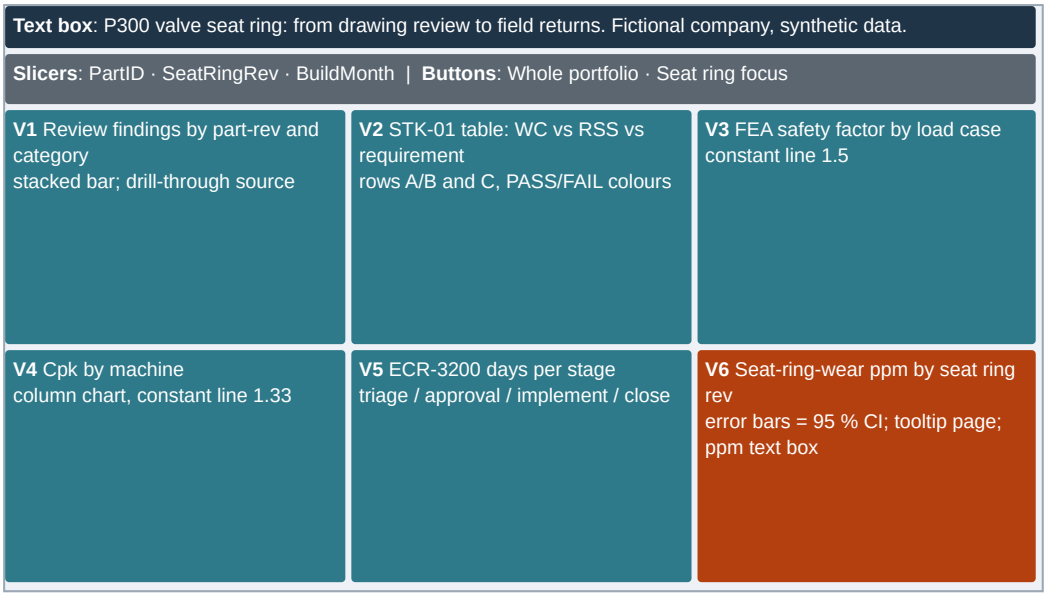
The valve seat ring story dashboard

One story page, one drill-through page, one tooltip page, two bookmarks with buttons. Start file: Section11_Start.pbix (the finished S10 file); save as Engxtech_S11.

THE STORY

The P300 valve seat ring went from rev A to rev B to rev C. Rev B had the longest drawing delay (19 days late after 3 review rounds), one of its two grinders runs at Cpk 0.787, and rev B has the highest field wear rate. ECR-3200 tightened the seat ring OD tolerance and released rev C. The page lets a design reviewer follow that thread in one minute, and shows how sure the data lets us be: rev C is an early signal only.

Page layout



Page 'Seat ring story', 16:9 canvas 1280 x 720. Read left to right, top to bottom: design review, verification, strength, manufacturing, change, field.

Visuals (copy each from its page with Ctrl+C, Ctrl+V)

#	Visual	Fields and measures	Format and interactions
V1	Stacked bar chart (copied from the Findings page, S04)	Y: Drawings[PartRev]; X: Finding Count; Legend: ReviewFindings[Category]	Sort by Finding Count descending. Carries Drawings[PartRev], so it can drill through. Part slicer: Filter.
V2	Table (copied from the Stacks page, S05)	SeatRingBuild[Build] first; Req Min, Req Max, WC Min, WC Max, WC Result, RSS Min, RSS Max, RSS Result; visual filter Stacks[StackID] = STK-01	Background colour fx = WC Colour / RSS Colour. Part slicer: None.
V3	Clustered bar chart (FEA & Cpk page, S05)	Y: FEA[Case]; X: Min SF	Constant line 1.5; bar colour fx = SF Colour; sort ascending. Part slicer: Filter.
V4	Clustered column chart (FEA & Cpk page, S05)	X: ManufacturingCpk[MachineID]; Y: Min Cpk	Constant line 1.33; 3 decimals; colour fx = Cpk Colour. Part slicer: Filter.
V5	Stacked bar chart (Changes page, S08)	Y: Changes[ChangeID] (visual filter ECR-3200 only); X: TriageDays, ApprovalDays, ImplementDays, CloseDays	Title: ECR-3200 days per stage (123 d in total). Part slicer: None.
V6	Clustered column chart (Validation & field page, S09)	X: SeatRingRevs[SeatRingRev]; Y: Field ppm	Error bars CI Low ppm / CI High ppm; tooltip = page 'ppm counts'. All three slicers: Filter (the only visual the SeatRingRev and BuildMonth slicers filter).

Measures used

No new data and no new measures: everything is in the model from S04 to S09 (in _Measures): Finding Count, Major Findings, Findings Cum %, Req Min/Max, WC Min/Max, RSS Min/Max, WC Result, RSS Result, WC Colour, RSS Colour, Min SF, SF Colour, Min Cpk, Cpk Colour, Wear Failures 90d, Matured Units, Field ppm, CI Low ppm, CI High ppm; columns FEA[Case], Drawings[DaysLate], Changes[TriageDays], [ApprovalDays], [ImplementDays], [CloseDays].

Edit interactions (Format > Edit interactions)

- Part slicer: Filter on V1, V3, V4; None on V2, V5, V6.
- SeatRingRev slicer and BuildMonth slicer: None on V1 to V5; Filter on V6.

Drill-through page 'Drawing detail'

- Build visual > Drill through well: Drawings[PartRev]; Keep all filters: On; the Back button appears top left.
- Table: Drawings[DrawingNo], [Sheet], [PlannedRelease], [ActualRelease], [DaysLate], [ReviewRounds]. Table: ReviewFindings[FindingID], [Category], [Severity], [OpenedDate], [ClosedDate], sorted by OpenedDate. Cards: Finding Count, Major Findings.
- The review findings Pareto (Category, Finding Count, Findings Cum %) copied from the Findings page: on this page it shows the categories of the picked drawing set.
- Model path: Drawings[PartRev] *:1 Revisions[PartRev]; ReviewFindings[DrawingNo] *:1 Drawings[DrawingNo].

Tooltip page 'ppm counts'

- Format page > Page information: Allow use as tooltip On; Canvas settings: Type Tooltip (320 x 240).
- Cards: Wear Failures 90d, Matured Units, Field ppm, CI Low ppm, CI High ppm; text box 'Matured build months only'. Optional: hide the page.
- On V6: Format visual > General > Tooltips: On; Type Report page; Page ppm counts.

Bookmarks (View > Bookmarks) and buttons

Bookmark	State to capture
Whole portfolio	All three slicers cleared: every part shows.
Seat ring focus	Part slicer = P300.

Insert > Buttons > Blank, text on, Action type Bookmark: one button per bookmark, placed in the slicer strip. After any change to the page, right-click each bookmark > Update.

Acceptance checklist

☐ The title text box says 'Fictional company, synthetic data'; nothing suggests the numbers come from a real company.

☐ Part slicer = P300: V1 shows P300-A 5, P300-B 9, P300-C 1 (15 findings); V3 shows P300-B 1.53 and P300-C 1.78; V4 shows GRD-01 1.848 and GRD-02 0.787.

☐ V2 row A/B: WC -0.001..0.101 (fail) mm; row C: WC 0.006..0.094 (fail) and RSS 0.0298..0.0702 (pass) mm. STK-01 is a diametral clearance; its upper end is overstated by the one-sided geometric terms.

☐ V3: only P200-B Drop 1 m 1.40 is below the 1.5 line. The P300 seating load change (1.53 to 1.78) is not presented as an effect of ECR-3200.

☐ V4: GRD-02 = 0.787 is the lowest; 3 machines are below 1.33 (CNC-M03 1.329 included).

☐ V5: ECR-3200 123 d / 57 d (cycle / approval stage); median closed cycle 47.0 d, p90 83.7 d for comparison.

☐ V6: rev A 2255 ppm (8/3548); 95 % CI 973-4443 ppm; rev B 2892 ppm (81/28007); 95 % CI 2297-3595 ppm; rev C 1710 ppm (4/2339); 95 % CI 466-4379 ppm; matured build months only (Mar 2025 to Jun 2026).

☐ The ppm text box gives the definition; any caption calls rev C an early signal (one matured build month, interval includes rev B's rate and overlaps rev A). Never 'fixed'.

☐ Drill-through: right-click the P300-B bar in V1 > Drill through > Drawing detail shows DRW-1008 / 19 d / 3 (drawing / days late / rounds), major findings / Tolerance 4 / 6 of 9; the Back button returns (Ctrl+click in Desktop).

☐ Tooltip: hovering the rev C column in V6 shows the 'ppm counts' page: 4 · 2,339 · 1,710 · 466 · 4,379.

☐ Bookmarks: Whole portfolio clears the slicers; Seat ring focus sets Part = P300; each button runs its bookmark (Ctrl+click in Desktop).

☐ Every title names the measure and the unit (mm, d, ppm) and, where there is one, the target; mass and cost are never on the same axis.